1 Calculation

1.1 From extraction metrics to routing outcomes

The simulation requires ticket-level routing outcomes rather than only precision/recall. Let N denote the total number of evaluated tickets, n_{dec} the number of tickets for which the automation emits a routing decision after thresholding, and TP the number of correct automated decisions. Under the TIIR deployment assumption of *at most one routing decision per ticket*, the metrics map as follows:

$$P = \frac{TP}{n_{\text{dec}}}, \tag{1}$$

$$R = \frac{TP}{N}, \tag{2}$$

$$C_{\text{dec}} = \frac{n_{\text{dec}}}{N}. \tag{3}$$

Therefore,

$$C_{\text{dec}} = \frac{R}{P}, \tag{4}$$

and the three simulation outcome probabilities become

$$p_{\text{correct}} = R, \tag{5}$$

$$p_{\text{auto-misroute}} = C_{\text{dec}} - R = R\left(\frac{1}{P} - 1\right), \tag{6}$$

$$p_{\text{abstain}} = 1 - C_{\text{dec}} = 1 - \frac{R}{P}. \tag{7}$$

For the TIIR model this yields

$$C_{\text{dec}} = \frac{0.4412}{0.8155} = 0.5410, \tag{8}$$

$$p_{\text{correct}} = 0.4412, \tag{9}$$

$$p_{\text{auto-misroute}} = 0.0998, \tag{10}$$

$$p_{\text{abstain}} = 0.4590. \tag{11}$$

1.2 Component-Level Calibration

The deployed CPE extraction model achieves precision $P = 0.8155$ and recall $R = 0.4412$ on vendor-product identity matching. To translate these metrics into simulation outcomes, we use the single-decision-per-ticket assumption formalized in Equation 4. This yields:

$$C_{\text{dec}} = \frac{R}{P} = 0.5410, \tag{12}$$

$$p_{\text{correct}} = R = 0.4412, \tag{13}$$

$$p_{\text{auto-misroute}} = C_{\text{dec}} - R = 0.0998, \tag{14}$$

$$p_{\text{abstain}} = 1 - C_{\text{dec}} = 0.4590. \tag{15}$$

Thus, in the simulation, 44.12% of tickets are correctly routed directly by TIIR, 9.98% are initially misrouted by the automation and subsequently require human rerouting, and 45.90% are abstained and sent directly to the manual routing process. In the manual baseline, there is no abstention state.

1.3 System-Level Mean TTCA Model

Primary empirical anchors. The manual routing service-time distribution is calibrated from the "Open → Assignment" transition reported by Trinkenreich et al., with mean 109.3 minutes, median 51.5 minutes, and maximum 26.3 hours. We model this stage with a capped log-normal distribution because the source provides both mean and median and explicitly identifies this transition as the main process bottleneck.

Misrouting is calibrated from Chen et al., who study incident triage on 20 real-world online service systems and report subject-level reassignment rates from 4.11% to 91.58% together with an average 10.16× increase in incident-triage time when reassignment occurs. In Chen et al., reassignment means that the incident was not assigned to the correct team on the first attempt. Because the paper does not publish ticket counts per subject, a globally weighted reassignment rate cannot be reconstructed. We therefore use the unweighted mean of the 20 published subject-level reassignment rates in Figure 2 of Chen et al. as a transparent baseline for the per-attempt human misrouting probability:

$$p_{\text{mis,human}} = 0.218155 \approx 0.2182. \tag{16}$$

Manual routing baseline and rerouting delay calibration. Manual incident routing is modeled as an iterative process where assignment attempts continue until the correct team is reached. Under a constant per-attempt human misrouting probability $p_{\text{mis,human}}$, the number of human routing attempts K follows a geometric distribution with success probability $1 - p_{\text{mis,human}}$:

$$\Pr(K = k) = p_{\text{mis,human}}^{k-1}(1 - p_{\text{mis,human}}), \quad k = 1, 2, \ldots \tag{17}$$

This captures the repeated ownership-search behavior observed in manual incident triage.

Let S_{human} denote the manual routing time per attempt with mean $\mu_S =$ 109.3 minutes, calibrated from the "Open → Assignment" transition reported by Trinkenreich et al.. Conditioned on at least one reassignment ($K \geq 2$),

$$\mathbb{E}[K \mid K \geq 2] = \frac{2 - p_{\text{mis,human}}}{1 - p_{\text{mis,human}}}. \tag{18}$$

We calibrate the inter-handoff reroute delay t_{reroute} such that the expected MTTCA of reassigned tickets reproduces the 10.16× inflation reported by Chen et al.:

$$\mathbb{E}[T \mid K \geq 2] = \mathbb{E}[K \mid K \geq 2] \cdot \mu_S + (\mathbb{E}[K \mid K \geq 2] - 1) \cdot t_{\text{reroute}} = 10.16 \cdot \mu_S. \tag{19}$$

Substituting $\mu_S = 109.3$ and $p_{\text{mis,human}} = 0.2182$ yields

$$t_{\text{reroute}} = 673.5 \text{ minutes } \approx 11.22 \text{ hours.} \tag{20}$$

This value represents the expected additional delay introduced by one misrouting cycle, including wrong-team ownership, queue re-entry, handoff latency, and renewed human triage.

Expected manual routing delay. Because manual routing may require multiple assignment attempts until the correct team is reached, the expected number of human routing attempts is

$$\mathbb{E}[K] = \frac{1}{1 - p_{\text{mis,human}}}, \tag{21}$$

and the expected number of human reassignment events is

$$\mathbb{E}[K-1] = \frac{p_{\text{mis,human}}}{1 - p_{\text{mis,human}}}. \tag{22}$$

Using these expectations, the mean time to correct assignment in the manual baseline becomes

$$MTTCA_{\text{manual}} = \mathbb{E}[K] \cdot \mu_S + \mathbb{E}[K-1] \cdot t_{\text{reroute}}. \tag{23}$$

Substituting the calibrated parameters yields

$$MTTCA_{\text{manual}} \approx 327.7 \text{ minutes.} \tag{24}$$

TIIR scenario and fairness of comparison. TIIR replaces only the first routing decision. For each arriving ticket, the automation first produces one of three outcomes:

1. **correct** with probability $p_{\text{correct}} = 0.4412$,
2. **auto-misroute** with probability $p_{\text{auto-misroute}} = 0.0998$,
3. **abstain** with probability $p_{\text{abstain}} = 0.4590$.

A correct automated routing exits the system after the automation time $t_{\text{auto}} + t_{\text{handoff}}$. If TIIR abstains, the ticket enters the same manual routing process as in the baseline system, i.e., the subsequent human ownership search remains geometric. If TIIR misroutes a ticket, the ticket first incurs one wrong automated handoff and then continues through the same manual routing process as in the baseline. Thus, TIIR does not artificially eliminate iterative misrouting cascades after handover; it changes only the first routing decision.

Queueing model. We simulate a routing queue with $c = 5$ parallel human routers and Poisson arrivals at rate λ tickets/hour. Each manual routing attempt draws a capped log-normal service time from the Trinkenreich calibration. If a human assignment is wrong, the ticket incurs a reroute delay t_{reroute} and re-enters the queue. Under TIIR, only abstained and auto-misrouted tickets enter the human queue; once in the human queue, subsequent routing follows the same manual process as in the baseline. We cap the number of reassignments only to prevent infinite loops in simulation; the cap is a numerical safeguard rather than a modeling claim.

```
Incident
|
+- TIIR correct -> done
|
+- TIIR abstain -> manual routing (geometric)
|
+- TIIR misroute -> reroute -> manual routing (geometric)
```

Fig. 1: Routing logic in the TIIR scenario. TIIR replaces only the first routing decision; abstained and auto-misrouted tickets continue through the same manual routing process as in the baseline.

Automation time. The default TIIR routing stage is modeled as $t_{\text{auto}} + t_{\text{handoff}} =$ 75 seconds (60 seconds inference + 15 seconds handoff), reflecting the architecture of a fast retrieval-backed inference pipeline rather than a human-reviewed support-desk assistant. We do *not* replace this value with the 4.5-minute U.S. Navy figure, because that external source measures a different workflow: ML-assisted ticket assignment with service-desk review, not the TIIR free-text→CPE→IAM path. Instead, the Navy value is exposed as an optional comparator preset in the released simulation script.

Contextual benchmark presets. To make the time ranges from the source corpus reproducible, the simulation code includes optional presets for the following empirical anchors:

- **SEKE / Organization A**: mean 109.3 min, median 51.5 min, cap 26.3 h,
- **Navy manual assignment**: median 17 min, ML-assisted 4.5 min,
- **Freshservice IT benchmark**: AFAT 15.80 h,
- **Freshservice ESM benchmark**: AFAT 28.93 h,
- **Freshservice fast channel**: AFAT 4.16 h for MS Teams/Slack,
- **Freshservice company-size anchors**: 14.22 h for organizations with fewer than 250 employees and 20.465 h for organizations with more than 1,000 employees.

These presets are not all equally suitable for the primary paper claims. The SEKE+Chen combination is the main calibration because it directly models *assignment to the correct team*; the Freshservice presets are retained for sensitivity analysis and contextual comparison only.

1.4 Outputs and Statistical Treatment

For each scenario, we report:

1. mean MTTCA,
2. median MTTCA,
3. run-level 95th percentile MTTCA,

4. SLA violation rate $\Pr[\text{MTTCA} > T_{\text{SLA}}]$,
5. human reassignment rate and human effort minutes.

The 95th percentile is computed per run and summarized with a t-interval over the run-level percentiles. Proportion metrics use Wilson intervals. This matches the released code exactly and avoids the earlier bootstrap mismatch.

SLA threshold. We use $T_{\text{SLA}} = 60$ minutes as an operational tail-latency threshold for the routing layer. This is an illustrative threshold for comparing scenarios, not a universal industry constant. The key use of the SLA metric is to expose how often routing alone would violate a one-hour internal target under different load and misrouting regimes.

1.5 Analytic pre-check before simulation

Before executing DES, the parameterization already reveals two important effects.

First, the lower-bound utilization of the manual routing stage is

$$\rho_{\text{manual,LB}} = \frac{\lambda \cdot \mathbb{E}[S_{\text{human}}]}{c \cdot 60}. \tag{25}$$

At the default $\lambda = 2$ tickets/hour and $c = 5$, this gives

$$\rho_{\text{manual,LB}} = \frac{2 \cdot 109.3}{5 \cdot 60} = 0.7287. \tag{26}$$

Second, under TIIR, only abstained and auto-misrouted tickets consume human routing capacity in the first approximation:

$$\rho_{\text{tiir,LB}} = \frac{\lambda \cdot (p_{\text{abstain}} + p_{\text{auto-misroute}}) \cdot \mathbb{E}[S_{\text{human}}]}{c \cdot 60} = \frac{2 \cdot 0.5588 \cdot 109.3}{5 \cdot 60} = 0.4073. \tag{27}$$

These lower bounds ignore reassignment loops; the DES accounts for them explicitly. Nevertheless, even this conservative pre-check already shows the central mechanism: TIIR primarily helps by removing a large fraction of tickets from the human routing queue ... and by reducing the number of tickets that enter repeated manual ownership search.

Table 1: Simulation configuration and primary calibration inputs used in v6. This table makes the DES setup explicit before any results are interpreted.

Category	Parameter	Value
Workload	Routers c	5
Workload	Tickets per run	200
Workload	Replications per scenario	200
Workload	Baseline arrival rate λ	2.0 tickets/hour
Workload	Load sweep	$\{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ tickets/hour
Workload	Routing SLA threshold	60 minutes
Workload	Maximum reassignments	6
Calibration	Timing preset	`seke_orgA`
Calibration	Manual mean routing time	109.3 minutes
Calibration	Manual median routing time	51.5 minutes
Calibration	Manual routing cap	1578.0 minutes (26.3 h)
Calibration	Human misrouting probability $p_{\text{mis,human}}$	0.2182
Calibration	Reassignment inflation factor	10.16×
Calibration	Derived reroute delay t_{reroute}	673.47 minutes
TIIR	Precision P	0.8155
TIIR	Recall R	0.4412
TIIR	Decision coverage $C_{\text{dec}} = R/P$	0.5410
TIIR	Auto-routing time $t_{\text{auto}} + t_{\text{handoff}}$	1 minute (60 s)

Table 2: Load sweep summary over $\lambda \in \{0.5, \ldots, 4.0\}$ tickets/hour. For each λ, the table reports the analytic lower-bound utilizations and the DES outcomes. Mean TTCA entries are formatted as manual / TIIR.

λ	ρ_{manual}	ρ_{TIIR}	Mean TTCA (min)
0.5	0.182	0.102	323.9 / 260.7
1.0	0.364	0.204	330.0 / 257.0
1.5	0.546	0.305	355.7 / 257.3
2.0	0.729	0.407	496.1 / 268.0
2.5	0.911	0.509	691.4 / 280.0
3.0	1.093	0.611	922.7 / 293.8
3.5	1.275	0.713	1113.8 / 332.0
4.0	1.457	0.814	1308.9 / 387.4